

Landing Zone Architecture: Replicating Palantir's Perimeter on GCP

Palantir operates as an implicit, air-gapped security perimeter. Migrating Palantir workloads (BigQuery, Cloud Run, Dataproc) into a flat GCP project without a foundational Landing Zone violates enterprise security standards and guarantees an audit failure.

This document defines the strict **GCP Enterprise Landing Zone** required to safely host the migrated Palantir workloads, utilizing a Shared VPC and VPC Service Controls (VPC-SC).

1. Resource Hierarchy

Palantir uses "Spaces" to logically separate lines of business. In GCP, we map this to the formal Folder and Project hierarchy to ensure isolation and billing attribution.

Palantir Concept	GCP Implementation	Description
Organization	GCP Organization Node	The root of the enterprise tree (e.g., dhl.com).
Spaces	GCP Folders	Logical grouping (e.g., Folder: Logistics , Folder: Finance).
Projects / Use Cases	GCP Projects	We enforce a strict Hub-and-Spoke model. One Host Project for networking, and multiple Service Projects (Data, Apps, Analytics) for workloads.

2. Identity & Access Management (IAM)

Palantir utilizes proprietary roles (Owner, Editor, Viewer) deeply tied to their ontology.

Palantir Role	GCP IAM & Data Controls	Description
Platform Administrator	roles/owner (Restricted)	Limited to emergency break-glass accounts.
Data Steward	roles/datacatalog.admin + roles/bigquery.admin	Governs the taxonomy and policy tags.
Ontology Editor	roles/bigquery.dataEditor	Can mutate tables; often restricted to the Cloud Run CQRS Service Accounts.
Analyst (Viewer)	roles/bigquery.dataViewer	Read-only access, heavily filtered by Dataplex column/row-level security.

3. Networking: The Shared VPC

A standalone VPC does not scale for an enterprise migration. We utilize a **Shared VPC (Hub-and-Spoke)** model.

- **Host Project (`prj-net-hub`):** Owns the Shared VPC, subnets, firewall rules, and Cloud NAT.
- **Service Projects (`prj-data-logistics`, `prj-apps-actions`):** Consume subnets from the Host Project.
- **Egress Control:** Cloud NAT is configured in the Host Project. All egress traffic routes through a centralized inspection point.
- **B2B Extensibility:** Private Service Connect (PSC) is used to securely expose Cloud Run APIs (the migrated Workshop Actions) to external partners without exposing the internal VPC.

4. Security Perimeters: VPC Service Controls (VPC-SC)

Palantir's "walled garden" prevents data exfiltration by default. In GCP, we enforce this contractually using **VPC Service Controls**.

- **The Perimeter:** A VPC-SC perimeter is drawn around all data-hosting Service Projects.
- **Protected APIs:** `bigquery.googleapis.com`, `storage.googleapis.com`, `pubsub.googleapis.com`, `dataproc.googleapis.com`.
- **Exfiltration Prevention:** Even if an analyst holds `roles/bigquery.dataViewer`, they cannot run a `bq extract` command to a GCS bucket outside the perimeter.
- **Perimeter Bridging:** For CI/CD pipelines (GitHub Actions/Cloud Build) to deploy code, explicit ingress/egress rules are configured on the perimeter to allow strictly defined deployment traffic.

[!WARNING] VPC-SC misconfigurations are the #1 cause of pipeline failures during migration. The Landing Zone infrastructure-as-code must be validated in a sandbox environment before applying to production.